PACKAGE INSERT

Remifentanil Reig Jofre 5 mg powder for concentrate solution for injection or infusion

1 vial contains 5 mg Remifentanil (as Remifentanil hydrochloride).

PRODUCT DESCRIPTION

Parenteral powder for solution for injection or infusion (lyophilized powder)
White or almost white powder.
The appearance of the solution is clear, colourless and practically free from particulate material.
Remifentanil has been shown to be compatible with the following intravenous fluids :

Water for Injections
Glucose 50 mg/ml (5%) solution for injection
Glucose 50 mg/ml (5%) and Sodium Chloride 9 mg/ml (0.9%) solution for injection
Sodium Chloride 9 mg/ml (0.9%) solution for injection
Sodium Chloride 4.5 mg/ml (0.45%) solution for injection

After dilution, the solution should be inspected visually to ensure it is clear, colourless, practically free from particulate matter and the container is undamaged.

THERAPEUTIC INDICATIONS

Remifentanil is indicated as an analgesic agent for use during induction and/or maintenance of general anaesthesia during surgical procedures including cardiac surgery, and also fir continuation of analgesia into the immediate post-operative period under close supervision, during transition to longer acting analgesia.

Remifentanil is indicated for provision of analgesia and sedation in mechanically ventilated intensive care patients.

POSOLOGY AND METHOD OF ADMINISTRATION

Remifentanil shall be administered only in a setting fully equipped for the monitoring and support of respiratory and cardiovascular function, and by persons specifically trained in the use of anaesthetic drugs and the recognition and management of the expected adverse effects of potent opioids, including respiratory and cardiac resuscitation. Such training must include the establishment and maintenance of a patent airway and assisted ventilation.

Continuous infusions of Remifentanil must be administered by a calibrated infusion device into a fast flowing IV line or via a dedicated IV line. This infusion line should be connected at, or close to, the venous cannula and primed to minimise the potential dead space (see the following additional information, including tables with examples of infusion rates by body weight to help titrate Remifentanil to the patient's anaesthetic needs).

Care should be taken to avoid obstruction or disconnection of infusion lines and to adequately clear the lines to remove residual Remifentanil after use (see section **Special warnings and precautions for use**).

Remifentanil is for intravenous use only and must not be administered by epidural or intrathecal injection (see section **Contraindications**).

Remifentanil for injection should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user after reconstitution and further dilution to 20 to 250 micrograms/ml (50 micrograms/ml is the recommended dilution for adults and 20 to 25 micrograms/ml for paediatric patients aged 1 year and over) with one of the following i.v. fluids listed below:

Water for Injections
Glucose 50 mg/ml (5%) solution for injection
Glucose 50 mg/ml (5%) and Sodium Chloride 9 mg/ml (0.9%) solution for injection
Sodium Chloride 9 mg/ml (0.9%) solution for injection
Sodium Chloride 4.5 mg/ml (0.45%) solution for injection

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.
(see **Special precautions for disposal and other handling** for additional information, including tables to help titrate Remifentanil to the patient's anaesthetic needs).

The administration of Remifentanil during general anaesthesia must be individualised based on the patient's response. It is not recommended for use as the sole agent in general anaesthesia.

GENERAL ANAESTHESIA

- **Adults**

The following table summarizes the starting injection/infusion rates and dose range.

DOSING GUIDELINES FOR ADULTS:

Indication	Bolus Infusion of remifentanil (micrograms/kg)	Continuous Infusion of remifentanil (micrograms/kg/h)	
		Starting Rate	Range
Induction of anaesthesia in ventilated patients	1 (give over not less than 30 seconds)	0.5 to 1	-
Maintenance of anaesthesia in ventilated patients			
Nitrous oxide (66%)	0.5 to 1	0.4	0.1 to 2
Isoflurane(starting dose 0.5 MAC)	0.5 to 1	0.25	0.05 to 2
Propofol(Starting dose100 micrograms/kg/min)	0.5 to 1	0.25	0.05 to 2

Spontaneous ventilation anaesthesia	Not recommended	0.04	0.025 to 0.1
Continuation of analgesia intothe immediate post-operative period	Not recommended	0.1	0.025 to 0.2

When given by bolus injection Remifentanil should be administered over not less than 30 seconds.
At the doses recommended above, Remifentanil significantly reduces the amount of hypnotic agent required to maintain anaesthesia. Therefore, isoflurane and propofol should be administered as recommended above to avoid excessive depth of anaesthesia (see Concomitant medication in this section). No data are available for dosage recommendations for simultaneous use of other hypnotics with Remifentanil.

Induction of anaesthesia
Remifentanil should be administered with a standard dose of hypnotic agent, such as propofol, thiopental, or isoflurane, for the induction of anaesthesia.
Remifentanil can be administered at an infusion rate of 0.5 to 1 micrograms/kg/min, with or without an initial bolus injection of 1 micrograms/kg given over not less than 30 seconds. If endotracheal intubation is to occur more than 8 to 10 minutes after the start of the infusion of Remifentanil, then a bolus injection is not necessary.

Maintenance of anaesthesia
After endotracheal intubation, the infusion rate of Remifentanil should be decreased, according to anaesthetic technique, as indicated in the above table.
Due to the fast onset and short duration of action of Remifentanil, the rate of administration during anaesthesia can be titrated upward in 25% to 100% increments or downward in 25% to 50% decrements, every 2 to 5 minutes to attain the desired level of μ-opioid response. In response to light anaesthesia, supplemental bolus injections may be administered every 2 to 5 minutes.

Anaesthesia in spontaneously breathing anaesthetised patients with a secured airway (e.g. laryngeal mask anaesthesia):
In spontaneously breathing anaesthetised patients with a secured airway respiratory depression is likely to occur. Special care is needed to adjust the dose to the patient requirements and ventilatory support may be required. The recommended starting infusion rate for supplemental analgesia in spontaneously breathing anaesthetised patients is 0.04 micrograms/kg/min with titration to effect. A range of infusion rates from 0.025 to 0.1 micrograms/kg/min has been studied.Bolus injections are not recommended in spontaneously breathing anaesthetised patients.

Continuation into the immediate post-operative period
In the event that longer acting analgesia has not been established prior to the end of surgery, Remifentanil may need to be continued to maintain analgesia during the immediate post-operative period until longer acting analgesia has reached its maximum effect.

In ventilated patients, the infusion rate should continue to be titrated to effect.
In patients who are breathing spontaneously, the infusion rate of Remifentanil should initially be decreased to a rate of 0.1 micrograms/kg/min. The infusion rate may then be increased or decreased by not greater than 0.025 micrograms/kg/min every five minutes, to balance the patient's level of analgesia and respiratory rate.

Remifentanil should only be used in a setting fully equipped for the monitoring and support of respiratory and cardiovascular function, under the close supervision of persons specifically trained in the recognition and management of the respiratory effects of potent opioids.

The use of bolus injections of Remifentanil to treat pain during the postoperative period is not recommended in patients who are breathing spontaneously.

Concomitant medication:
Remifentanil decreases the amounts or doses of inhaled anaesthetics, hypnotics and benzodiazepines required for anaesthesia (see section **Interaction with other medicinal products and other forms of interaction**).

Doses of the following agents used in anaesthesia: isoflurane, thiopentone, propofol and temazepam have been reduced by up to 75% when used concurrently with Remifentanil.

Guidelines for discontinuation:
Due to the very rapid offset of action of Remifentanil no residual opioid activity will be present within 5 to 10 minutes after discontinuation. For those patients undergoing surgical procedures where post-operative pain is anticipated, analgesics should be administered prior to or immediately following discontinuation of Remifentanil. Sufficient time must be allowed to reach the maximum effect of the longer acting analgesic. The choice of analgesic should be appropriate for the patient's surgical procedure and the level of post-operative care.

- **Paediatric patients (1 to12 years of age)**

Induction of anaesthesia
There are insufficient data to make a dosage recommendation.

Maintenance of anaesthesia

DOSING GUIDELINES FOR MAINTENANCE OF ANAESTHESIA PAEDIATRIC PATIENTS (1 to 12 years of age):

Concomitant Anaesthetic Agent	Bolus Infusion of Remifentanil (micrograms/kg)	Continuous Infusion of Remifentanil (micrograms/kg/min)	
		Starting Rate	Typical Maintenance Rates
Nitrous oxide (70%)	1	0.4	0.4 to 3
Halothane (starting dose 0.3 MAC)	1	0.25	0.05 to 1.3
Sevoflurane	1	0.25	0.05 to 0.9

(starting dose 0.3 MAC)			
Isoflurane (starting dose 0.5 MAC)	1	0.25	0.06 to 0.9

When given by bolus infusion, Remifentanil should be administered over not less than 30 seconds. Surgery should not commence until at least 5 minutes after the start of Remifentanil infusion, if a simultaneous bolus dose has not been given. Paediatric patients should be monitored and the dose titrated to the depth of analgesia appropriate for the surgical procedure.

Concomitant medication
At the doses recommended above, Remifentanil significantly reduces the amount of hypnotic agent required to maintain anaesthesia. Therefore, isoflurane, halothane and sevoflurane should be administered as recommended above to avoid excessive depth of anaesthesia. No data are available for dosage recommendations for simultaneous use of other hypnotics other than those listed in the table with Remifentanil (see in **Posology and method of administration***: General Anaesthesia / Adults – Concomitant medication).*

Guidelines for discontinuation
Following discontinuation of the infusion, the offset of analgesic effect of Remifentanil is rapid and similar to that seen in adult patients. Appropriate post-operative analgesic requirements should be anticipated and implemented (see Dosage and Administration- General Anaesthesia- Adults- Guidelines for discontinuation).

- **Neonates / Infants (Aged less than 1 year)**

The pharmacokinetic profile of Remifentanil in neonates/infants (aged less than 1 year) is comparable to that seen in adults after correction for body weight differences. However, there are insufficient clinical data to make dosage recommendations for this age group.

CARDIAC ANAESTHESIA

- **Adults**

DOSING GUIDELINES FOR CARDIAC ANAESTHESIA:

Indication	Bolus Injection (µg/kg)	Continuous Infusion (µg/kg/min)	
		Starting Rate	Typical Infusion Rates
Intubation	Not recommended	1	-
Maintenance of anaesthesia - Isoflurane (starting dose 0.4 MAC)	0.5-1	1	0.003 to 4
- Propofol (starting dose 50 µg/kg/min)	0.5-1	1	0.01 to 4.3

Continuation of postoperative analgesia, prior to extubation	Not recommended	1	0 to 1

Induction period of anaesthesia
After administration of hypnotic to achieve loss of consciousness, Remifentanil should be administered at an initial infusion rate of 1µg/kg/min. The use of bolus injections of Remifentanil during induction in cardiac surgical patients is not recommended. Endotracheal intubation should not occur until at least 5 minutes after the start of the infusion.

Maintenance period of anaesthesia
After endotracheal intubation the infusion rate of Remifentanil should be titrated according to patient need. Supplemental slow bolus doses may also be given as required. High risk cardiac patients, such as those with poor ventricular function, should be administered a maximum bolus dose of 0.5 micrograms/kg.

These dosing recommendations also apply during hypothermic cardiopulmonary bypass (see s*ection* ***Pharmacokinetic properties***).

Concomitant medication
At the doses recommended above, Remifentanil significantly reduces the amount of hypnotic agent required to maintain anaesthesia. Therefore, isoflurane and propofol should be administered as recommended above to avoid excessive depth of anaesthesia. No data are available for dosage recommendations for simultaneous use of other hypnotics with Remifentanil (see *section* ***Posology and method of administration****:* G*eneral Anaethesia / Adults/ Concomitant medication).*

Continuation of post-operative analgesia prior to extubation
It is recommended that the infusion of Remifentanil should be maintained at the final intra-operative rate during transfer of patients to the post-operative care area. Upon arrival into this area, the patient's level of analgesia and sedation should be closely monitored and the Remifentanil infusion rate adjusted to meet the individual patient's requirements.

Guidelines for discontinuation
Prior to discontinuation of Remifentanil, patients must be given alternative analgesic and sedative agents at a sufficient time in advance. The choice and dose of agent(s) should be appropriate for the patient's level of post-operative care (see Dosage and Administration-General Anaesthesia- Adults- Guidelines for discontinuation).

It is recommended that the Remifentanil infusion is discontinued by reducing the infusion rate by 25% decrements in at least 10-minute intervals until the infusion is discontinued. During weaning from the ventilator the Remifentanil infusion should not be increased and only down titration should occur, supplemented as required with alternative analgesics.

It is recommended that haemodynamic changes such as hypertension and tachycardia should be treated with alternative agents as appropriate.

- **Paediatric Patients**

There are insufficient data to make a dosage recommendation for use during cardiac surgery.

USE IN INTENSIVE CARE

- **Adults**

Remifentanil can be initially used alone for the provision of analgesia and sedation in mechanically ventilated intensive care patients.
It is recommended that Remifentanil is initiated at an infusion rate of 0.1 micrograms/kg/min to 0.15 micrograms/kg/min. The infusion rate should be titrated in increments of 0.025 micrograms/kg/min to achieve the desired level of analgesia and sedation. A period of at least 5 minutes should be allowed between dose adjustments. The level of analgesia and sedation should be carefully monitored, regularly reassessed and the Remifentanil infusion rate adjusted accordingly. If an infusion rate of 0.2 micrograms/kg/min is reached and the desired level of sedation is not achived, it is recommended that dosing with an appropriate sedative agent is initiated. The dose of sedative agent should be titrated to obtain the desired level of sedation. Further increases to the Remifentanil infusion rate in increments of 0.025 micrograms/kg/min may be made if additional analgesia is required.
Remifentanil has been studied in intensive care patients in well controlled clinical trials for up to 3 days. There are limited additional clinical trial data for longer durations.

The following table summarises the starting infusion rates and typical dose range for provision of analgesia in individual patient:

DOSING GUIDELINES FOR USE WITHIN THE INTENSIVE CARE SETTING

Continuous Infusion (micrograms/kg/min)	
Starting Rate	Range
0.1 to 0.15	0.006 to 0.74

Bolus doses of Remifentanil are not recommended in the intensive care setting.
The use of Remifentanil will reduce the dosage requirement of any concomitant sedative agents. Typical starting doses for sedative agents, if required, are given below.

RECOMMENDED STARTING DOSE OF SEDATIVE AGENTS, IF REQUIRED

Sedative Agents	Bolus (mg/kg)	Infusion (mg/kg/h)
Propofol	Up to 0.5	0.5
Midazolam	Up to 0.03	0.03

To allow separate titration of the respective agents, sedative agents should not be prepared as one mixture in the same infusion bag.

Additional analgesia for ventilated patients undergoing stimulating procedures
An increase in the existing Remifentanil infusion rate may be required to provide additional analgesic cover for ventilated patients undergoing stimulating and/or painful procedures such as endotracheal suctioning, wound dressing and physiotherapy. It is recommended that a Remifentanil infusion rate of at least 0.1 micrograms/kg/min should be maintained for at least 5 minutes prior to the start of the stimulating procedure.
Further dose adjustments may be made every 2 to 5 minutes in increments of 25% to 50% in anticipation of, or in response to, additional requirement for analgesia. A mean infusion rate of 0.25 micrograms/kg/min, maximum 0.75 micrograms/kg/min, has been administered for provision of additional analgesia during stimulating procedures.

Guidelines for discontinuation
Prior to discontinuation of Remifentanil, patients must be given alternative analgesic and sedative agents at a sufficient time in advance. The appropriate choice and dose of agent(s) should be anticipated and implemented.

In order to ensure a smooth emergence from a Remifentanil-based regimen it is recommended that the infusion rate of Remifentanil is titrated in stages to 0.1 micrograms/kg/min over a period up to 1 hour prior to extubation.

Following extubation, the infusion rate should be reduced by 25% decrements in at least 10 minutes intervals until the infusion is discontinued. During weaning from the ventilator the Remifentanil infusion should not be increased and only down titration should occur, supplemented as required with alternative analgesics.

- **Paediatric Intensive Care Patients**

There are no data available on use in paediatric patients

OTHER POPULATIONS

- **Elderly (over 65 years of age)**

GENERAL ANAESTHESIA:
The initial starting dose of Remifentanil administered to patients over 65 should be half the recommended adult dose and then titrated to individual patient need as an increased sensitivity to the pharmacological effects of Remifentanil has been seen in this patient population. This dose adjustment applies to use in all phases of anaesthesia including induction, maintenance, and immediate post-operative analgesia.

CARDIAC ANAESTHESIA:
No initial dose reduction is required (see also in *section **Posology and method of administration**: Cardiac anaesthesia-Dosing guidelines*).

INTENSIVE CARE:

No initial dose reduction is required (see also in *section* ***Posology and method of administration****: Use in Intensive Care*).

- Obese patients

It is recommended that for obese patients the dosage of Remifentanil should be reduced and based upon ideal body weight as the clearance and volume of distribution of remifentanil are better correlated with ideal body weight than actual body weight in this population.

- Renal impairment

No dosage adjustment relative to that used in healthy adults is necessary in renally impaired patients, including those undergoing renal replacement therapy, as the pharmacokinetic profile of Remifentanil is unchanged in this patient population.

- Hepatic impairment

No adjustment of the initial dose, relative to that used in healthy adults, is necessary as the pharmacokinetic profile of Remifentanil is unchanged in this patient population. However, patients with severe hepatic impairment may be slightly more sensitive to the respiratory depressant effects of remifentanil. These patients should be closely monitored and the dose of Remifentanil titrated to individual patient need.

- Neurosurgery

Limited clinical experience in patients undergoing neurosurgery has shown that no special dosage recommendations are required.

- **ASA III/IV patients**

GENERAL ANAESTHESIA

As the haemodynamic effects of potent opioids can be expected to be more pronounced in ASA III/IV patients, caution should be exercised in the administration of Remifentanil in this population. Initial dosage reduction and subsequent titration to effect is therefore recommended.

CARDIAC ANAESTHESIA

No initial dose reduction is required (see also in *section* ***Posology and method of administration****: Cardiac anaesthesia- Dosing guidelines*).

CONTRAINDICATIONS

As glycine is present in the formulation, Remifentanil is contra-indicated for epidural and intrathecal use. (see Pre-Clinical Safety Data).

Remifentanil is contra-indicated in patients with hypersensitivity to the active substance or other fentanyl analogues.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Remifentanil shall be administered only in a setting fully equipped for the monitoring and support of respiratory and cardiovascular function, and by persons specifically trained in the use of anaesthetic drugs and the recognition and management of the expected adverse effects of potent opioids, including respiratory and cardiac resuscitation. Such training must include the establishment and maintenance of a patent airway and assisted ventilation.

The use of Remifentanil in mechanically ventilated intensive care patients is not recommended for a duration of treatment greater than 3 days.

Rapid offset of action/ Transition to alternative analgesia:

Due to the very rapid offset of action of Remifentanil, no residual opioid activity will be present within 5-10 minutes after the discontinuation of Remifentanil. For those patients undergoing surgical procedures where post-operative pain is anticipated, analgesics should be administered prior to discontinuation of Remifentanil. The possibility of tolerance, hyperalgesia and associated haemodynamic changes should be considered when used in Intensive Care Unit. Prior to discontinuation of Remifentanil, patients must be given alternative analgesic and sedative agents. Sufficient time must be allowed to reach the therapeutic effect of the longer acting analgesic. The choice of agent(s), the dose and the time of administration should be planned in advance and individually tailored to be appropriate for the patient's surgical procedure and the level of post-operative care anticipated. When other opioid agents are administered as part of the regimen for transition to alternative analgesia, the benefit of providing adequate post-operative analgesia must always be balanced against the potential risk of respiratory depression with these agents.

Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Remifentanil with benzodiazepines.

Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

lf the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

lf the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

lf the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Remifentanil is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug lnteractions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Remifentanil with serotonergic drugs (See lnteractions with Other Medicaments).

This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See lnteractions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that.
Discontinue Remifentanil if serotonin syndrome is suspected.

Adrenal Insufficiency
Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. lf adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. lf adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction
Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

Discontinuation of treatment:

Symptoms following withdrawal of Remifentanil including tachycardia, hypertension and agitation have been reported infrequently upon abrupt cessation, particularly after prolonged administration of more than 3 days.
Where reported, re-introduction and tapering of the infusion has been beneficial. The use of Remifentanil in mechanically ventilated intensive care patients is not recommended for duration of treatment greater than 3 days.

Muscle rigidity - prevention and management:

At the doses recommended muscle rigidity may occur. As with other opioids, the incidence of muscle rigidity is related to the dose and rate of administration. Therefore, slow bolus injections should be administered over not less than 30 seconds.

Muscle rigidity induced by Remifentanil must be treated in the context of the patient's clinical condition with appropriate supporting measures.
Excessive muscle rigidity occurring during the induction of anaesthesia should be treated by the administration of a neuromuscular blocking agent and/or additional hypnotic agents. Muscle rigidity seen during the use of Remifentanil as an analgesic may be treated by stopping or decreasing the rate of administration of Remifentanil. Resolution of muscle rigidity after discontinuing the infusion of Remifentanil occurs within minutes. Alternatively an opioid antagonist may be administered; however this may reverse or attenuate the analgesic effect of Remifentanil.

Respiratory depression – prevention and management:

As with all potent opioids, profound analgesia is accompanied by marked respiratory depression. Therefore, Remifentanil should only be used in areas where facilities for monitoring and dealing with respiratory depression are available. Special care should be taken in patients with respiratory dysfunction.

The appearance of respiratory depression shall be managed appropriately, including decreasing the rate of infusion by 50%, or by a temporary discontinuation of the infusion. Unlike other fentanyl analogues, Remifentanil has not been shown to cause recurrent respiratory depression, even after prolonged administration. However, as many factors may affect post-operative recovery it is important to ensure that full consciousness and adequate spontaneous ventilation are achieved before the patient is discharged from the recovery area.

Theoretically, the risk of late depression should be absent with Remifentanil, however in the presence of confounding factors (inadvertent administration of bolus doses (see section below) and administration of concomitant longer acting opioids), respiratory depression occurring up to 50 minutes after discontinuation of infusion has been reported.

Cardiovascular effects:
The risk of cardiovascular effects such as hypotension and bradycardia, which may rarely lead to asystole/cardiac arrest (see section **Interaction with other medicinal products and other forms of interaction** and **Undesirable effects**) may be reduced by lowering the rate of infusion of Remifentanil or the dose of concurrent anaesthetics or by using IV fluids, vasopressor or anticholinergic agents as appropriate.

Debilitated, hypovolaemic, hypotensive and elderly patients may be more sensitive to the cardiovascular effects of Remifentanil.

Inadvertent administration:
A sufficient amount of Remifentanil may be present in the dead space of the IV line and/or cannula to cause respiratory depression, apnoea and/or muscle rigidity if the line is flushed with IV fluids or other drugs. This may be avoided by administering Remifentanil into a fast flowing IV line or via a dedicated IV line which is removed when Remifentanil is discontinued.

Neonates/infants (aged less than 1 year):
There is limited data available on use in neonates/infants under 1 year of age.

Drug abuse:
As with other opioids Remifentanil may produce dependency.

Interaction with other medicinal products and other forms of interaction

Remifentanil is not metabolised by plasmacholinesterase, therefore, interactions with drugs metabolised by this enzyme are not anticipated. As with other opioids Remifentanil, decreases the amounts or doses of inhaled and IV anaesthetics, and benzodiazepines required for anaesthesia (see *section* ***Posology and method of administration***). If doses of concomitantly administered CNS depressant drugs are not reduced patients may experience an increased incidence of adverse effects associated with these agents.

The cardiovascular effects of Remifentanil (hypotension and bradycardia), may be exacerbated in patients receiving concomitant cardiac depressant drugs, such as beta-blockers and calcium channel blocking agents.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.
The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration.
Opioids interact primarily at mu-receptors, and benzodiazepines interact at $GABA_A$ sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.
Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).
Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. lf concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Remifentanil if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

Fertility, pregnancy and lactation

There are no adequate and well-controlled studies in pregnant women.
Remifentanil should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

It is not known whether Remifentanil is excreted in human milk. However, because fentanyl analogues are excreted in human milk and Remifentanil-related material was found in rat milk after dosing with Remifentanil, nursing mothers should be advised to discontinue breast feeding for 24 hours following administration of Remifentanil.

Labour and delivery:

There are insufficient data to recommend Remifentanil for use during labour and Caesarean section. It is known that Remifentanil crosses the placental barrier and fentanyl analogues can cause respiratory depression in the child.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

After anaesthesia with Remifentanil the patient should not drive or operate machinery. The physician should decide when these activities may be resumed. It is advisable that the patient is accompanied when returning home and that alcoholic drink is avoided.

This medicine can impair cognitive function and can affect a patient's ability to drive safely. When prescribing this medicine, patients should be told:

- The medicine is likely to affect your ability to drive
- Do not drive until you know how the medicine affects you
- It is an offence to drive while under the influence of this medicine.
- However, you would not be committing an offence (called 'statutory defence') if:
    - The medicine has been prescribed to treat a medical or dental problem and
    - You have taken it according to the instructions given by the prescriber and in the information provided with the medicine and
    - It was not affecting your ability to drive safely

UNDESIRABLE EFFECTS

The most common undesirable effects associated with Remifentanil are direct extensions of μ-opioid agonist pharmacology. These adverse events resolve within minutes of discontinuing or decreasing the rate of Remifentanil administration.
Undesirable effects are listed below by system organ class and frequency.
Frequencies below are defined as very common (≥1/10), common (≥1/100 to <1/10), uncommon (≥1/1,000 and <1/100), are rare (≥1/10,000 to <1/1000), very rare (<1/10,000) and not known (cannot be estimated from the available data)

<u>Immune system disorders</u>

Rare: Allergic reactions including anaphylaxis have been reported in patients receiving Remifentanil in conjunction with one or more anaesthetic agents.

<u>Psychiatric disorders</u>
Not known: Drug dependence

<u>Nervous system disorders</u>
Very common: Skeletal muscle rigidity
Rare: Sedation (during recovery from general anaesthesia)
Not known: Convulsions

<u>Cardiac disorders</u>
Common: Bradycardia
Rare: Asystole/cardiac arrest, usually preceded by bradycadia, has been reported in patients receiving Remifentanil in conjunction with other anaesthetic agents.
Not known: Atrioventricular block

Vascular disorders
Very common: Hypotension
Common: Post-operative hypertension

Respiratory, thoracic and mediastinal disorders
Common: Acute respiratory depression, apnoea
Uncommon: Hypoxia

Gastrointestinal disorders
Very common: Nausea, vomiting
Uncommon: Constipation

Skin and subcutaneous tissue disorders
Common: Pruritus

General disorders and administration site conditions
Common: Post-operative shivering
Uncommon: Post-operative aches
Not known: Drug tolerance

Postmarketing Experience:
Serotonin syndrome (SeeWarnings and Precautions)
Adrenal insufficiency (SeeWarnings and Precautions)
Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamicpituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.
Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

Discontinuation of treatment
Symptoms following withdrawal of Remifentanil including tachycardia, hypertension and agitation have been reported infrequently upon abrupt cessation, particularly after prolonged administration of more than 3 days (see section **Special warnings and precautions for use**).

OVERDOSE AND TREATMENT
As with all potent opioid analgesics, overdose would be manifested by an extension of the pharmacologically predictable actions of Remifentanil. Due to the very short duration of action of Remifentanil, the potential for deleterious effects due to overdose are limited to the immediate time period following drug administration. Response to discontinuation of the drug is rapid, with return to baseline within ten minutes.

In the event of overdose or suspected overdose, take the following actions: discontinue administration of Remifentanil, maintain a patent airway, initiate assisted or controlled ventilation with oxygen and maintain adequate cardiovascular function. If depressed respiration is associated with muscle rigidity, a neuromuscular blocking agent may be required to facilitate assisted or controlled respiration.

Intravenous fluids and vasopressor for the treatment of hypotension and other supportive measures may be employed.

Intravenous administration of an opioid antagonist such as naloxone may be given as a specific antidote to manage severe respiratory depression and muscle rigidity. The duration of respiratory depression following overdose with Remifentanil is unlikely to exceed the duration of action of the opioid antagonist.

Pharmacodynamic properties

Pharmacotherapeutic group: Opioid anesthetics, ATC code: N01AH06

Remifentanil is a selective µ-opioid agonist with a rapid onset and very short duration of action. The µ-opioid activity of Remifentanil is antagonized by narcotic antagonists, such as naloxone.

PHARMACOKINETIC PROPERTIES

Following administration of the recommended doses of Remifentanil, the effective half-life is 3-10 minutes. The average clearance of Remifentanil in adolescents is 40ml/min/kg, the central volume of distribution is 100 ml/kg and the steady-state volume of distribution is 350 ml/kg. Blood concentrations of Remifentanil are proportional to the dose administered throughout the recommended dose range. For every 0.1µg/kg/min increase in infusion rate, the blood concentration of Remifentanil will rise 2.5ng/ml.

Remifentanil is approximately 70% bound to plasma proteins.

Metabolism:

Remifentanil is an esterase metabolised opioid that is susceptible to metabolism by non-specific blood and tissue esterases. The metabolism of Remifentanil results in the formation of a carboxylic acid metabolite which in dogs is 1/4600th as potent as Remifentanil. Studies in man indicate that all pharmacological activity is associated with the parent compound. The activity of this metabolite is therefore not of any clinical consequence. The halflife of the metabolite in healthy adults is 2 hours.

In patients with normal renal function, the time for 95% elimination of the primary metabolite of Remifentanil by the kidneys, is approximately 7 to 10 hours. Remifentanil is not a substrate for plasma cholinesterase.

Placental and milk transfer:

Placental transfer studies in rats and rabbits showed that pups are exposed to Remifentanil and/or its metabolites during growth and development.

Remifentanil-related material is transferred to the milk of lactating rats. In a human clinical trial, the concentration of Remifentanil in foetal blood was approximately 50% of that in maternal blood. The foetal arterio-venous ratio of Remifentanil concentrations was approximately 30%, suggesting metabolism of Remifentanil in the neonate.

Cardiac Anaesthesia:
The clearance of Remifentanil is reduced by approximately 20% during hypothermic (28°C) cardiopulmonary bypass. A decrease in body temperature lowers elimination clearance by 3% per degree centigrade.

Renal impairment:
In the clinical studies conducted to date, the rapid recovery from Remifentanil based analgesia appears unaffected by renal status.

The pharmacokinetics of Remifentanil are not significantly changed in patients with varying degrees of renal impairment even after administration for up to 3 days in the intensive care setting.

The clearance of the carboxylic acid metabolite is reduced in patients with renal impairment.

In intensive care patients with moderate/severe renal impairment, the concentration of the carboxylic acid metabolite is expected to reach approximately 100-fold the level of Remifentanil at steady-state. Clinical data demonstrates that accumulation of the metabolite does not result in clinically relevant μ-opioid effects even after administration of Remifentanil infusions for up to 3 days in these patients. There are no data available on the safety and pharmacokinetic profile of the metabolite following infusions of Remifentanil for durations greater than 3 days.

There is no evidence that Remifentanil is extracted during renal replacement therapy.

The carboxylic acid metabolite is extracted during haemodialysis by at least 30%.

Hepatic impairment:
The pharmacokinetics of Remifentanil are not changed in patients with severe hepatic impairment awaiting liver transplant, or during the anhepatic phase of liver transplant surgery. Patients with severe hepatic impairment may be slightly more sensitive to the respiratory depressant effects of Remifentanil.
These patients should be closely monitored and the dose of Remifentanil should be titrated to the individual patient need.

Paediatric patients:
The average clearance and steady state of volume of distribution of Remifentanil are increased in younger children and decline to adolescents values by age 17. The elimination half-life of Remifentanil in neonates is not significantly different from that of adolescents. Changes in analgesic effect after changes in infusion rate of Remifentanil should be rapid and similar to those seen in adolescents. The pharmacokinetics of the carboxylic acid metabolite in paediatric patients 2-17 years of age are similar to those seen in adults after correcting for differences in body weight.

Elderly:
The clearance of Remifentanil is slightly reduced (approximately 25%) in elderly patients>65 years) compared to young patients. The pharmacodynamic activity of Remifentanil increases with increasing age. Elderly patients have a Remifentanil EC50

for formation of delta waves on the electroencephalogram (EEG) that is 50% lower than young patients; therefore, the initial dose of Remifentanil should be reduced by 50% in elderly patients and then carefully titrated to meet the individual patient need.

INCOMPATIBILITIES

Remifentanil must not be mixed with other medicinal products except those mentioned in section **Special Precautions For Disposal.**

It should not be mixed with Lactated Ringer's Solution for Injection or Lactated Ringer's and Glucose 50 mg/ml (5%) Solution for Injection.

Remifentanil should not be mixed with propofol in the same intravenous admixture solution.
Administration of Remifentanil into the same intravenous line with blood/serum/plasma is not recommended. Non-specific esterases in blood products may lead to the hydrolysis of Remifentanil to its inactive metabolite.

SHELF LIFE

Vials:
Remifentanil 5mg: 3 years

Shelf life after reconstitution:
From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

Shelf life after dilution:
All mixtures with infusion fluids should be used immediately.

Special precautions for storage

Do not store above 25°C.
The reconstituted solution of Remifentanil should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. Remifentanil does not contain an antimicrobial preservative and thus care must be taken to assure the sterility of prepared solutions, reconstituted product should be used promptly, and any unused material discarded.

DOSAGE FORMS AND PACKAGING AVAILABLE

Glass vial (type I) with chlorobutyl rubber stopper and flip-off cap:

Remifentanil 5 mg Pack size of 5 x 5 mg/vial

SPECIAL PRECAUTIONS FOR DISPOSAL AND OTHER HANDLING

Remifentanil should be prepared for intravenous use by adding, as appropriate 1, 2 or 5ml of diluent to give a reconstituted solution with a concentration of approximately 1 mg/ml Remifentanil.

The reconstituted solution is clear, colourless, and practically free from particulate material.
After reconstitution the solution should be visually inspected on contaminations, colour or a defective container. The solution should be discarded, if any of these modifications should appear. Reconstituted product is for single use only. Residuals must be discarded.

Remifentanil should not be administered by manually-controlled infusion without further dilution to concentrations of 20 to 250 micrograms/ml (50 micrograms/ml is the recommended dilution for adults and 20 to 25 micrograms/ml in paediatric patients aged 1 year and over).

Remifentanil should not be administered by target-controlled infusion (TCI) without further dilution (20 to 50 micrograms/ml is the recommended dilution for TCI).

The dilution is dependent upon the technical capability of the infusion device and the anticipated requirements of the patient.

Remifentanil for injection should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user after reconstitution and further dilution to 20 to 250 micrograms/ml (50 micrograms/ml is the recommended dilution for adults and 20 to 25 micrograms/ml for paediatric patients aged 1 year and over) with one of the following i.v. fluids listed below:

Water for Injections
Glucose 50 mg/ml (5%) solution for injection
Glucose 50 mg/ml (5%) and Sodium Chloride 9 mg/ml (0.9%) solution for injection
Sodium Chloride 9 mg/ml (0.9%) solution for injection
Sodium Chloride 4.5 mg/ml (0.45%) solution for injection

After dilution, the solution should be inspected visually to ensure it is clear, colourless, practically free from particulate matter and the container is undamaged. Any solution where such defects are observed must be discarded.

Remifentanil has been shown to be compatible with the following i.v. fluids when administered into running i.v. infusion of:
− Lactated Ringer's injection
− Lactated Ringer's and 5% dextrose injection

Remifentanil has been shown to be compatible with propofol when administered into a running IV catheter.
Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

The following tables give guidelines for infusion rates of Remifentanil for manually-controlled infusion:

Table 1: Remifentanil – Injection Infusion Rates (ml/kg/h)

Drug Delivery Rate (µg/kg/min)	Infusion Rate (ml/kg/h) for Solution Concentrations of			
	20µg/ml 1mg/50ml	25µg/ml 1mg/40ml	50µg/ml 1mg/20ml	250µg/ml 10 mg/40ml
0.0125	0.038	0.03	0.015	Not recommended
0.025	0.075	0.06	0.03	Not recommended
0.05	0.15	0.12	0.06	0.012
0.075	0.23	0.18	0.09	0.018
0.1	0.3	0.24	0.12	0.024
0.15	0.45	0.36	0.18	0.036
0.2	0.6	0.48	0.24	0.048
0.25	0.75	0.6	0.3	0.06
0.5	1.5	1.2	0.6	0.12
0.75	2.25	1.8	0.9	0.18
1.0	3.0	2.4	1.2	0.24
1.25	3.75	3.0	1.5	0.3
1.5	4.5	3.6	1.8	0.36
1.75	5.25	4.2	2.1	0.42
2.0	6.0	4.8	2.4	0.48

Table 2: Remifentanil -Injection Infusion Rates (ml/h) for a 20µg/ml Solution

Infusion Rate (µg/kg/min)	Patient Weight (kg)						
	5	10	20	30	40	50	60
0.0125	0.188	0.375	0.75	1.125	1.5	1.875	2.25
0.025	0.375	0.75	1.5	2.25	3.0	3.75	4.5
0.05	0.75	1.5	3.0	4.5	6.0	7.5	9.0
0.075	1.125	2.25	4.5	6.75	9.0	11.25	13.5
0.1	1.5	3.0	6.0	9.0	12.0	15.0	18.0
0.15	2.25	4.5	9.0	13.5	18.0	22.5	27.0
0.2	3.0	6.0	12.0	18.0	24.0	30.0	36.0
0.25	3.75	7.5	15.0	22.5	30.0	37.5	45.0
0.3	4.5	9.0	18.0	27.0	36.0	45.0	54.0
0.35	5.25	10.5	21.0	31.5	42.0	52.5	63.0
0.4	6.0	12.0	24.0	36.0	48.0	60.0	72.0

Table 3: Remifentanil – Injection Infusion Rates (ml/h) for a 25 µg/ml Solution

Infusion Rate (µg/kg/min)	Patient Weight (kg)									
	10	20	30	40	50	60	70	80	90	100
0.0125	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0
0.025	0.6	1.2	1.8	2.4	3.0	3.6	4.2	4.8	5.4	6.0
0.05	1.2	2.4	3.6	4.8	6.0	7.2	8.4	9.6	10.8	12.0
0.075	1.8	3.6	5.4	7.2	9.0	10.8	12.6	14.4	16.2	18.0
0.1	2.4	4.8	7.2	9.6	12.0	14.4	16.8	19.2	21.6	24.0
0.15	3.6	7.2	10.8	14.4	18.0	21.6	25.2	28.8	32.4	36.0

0.2	4.8	9.6	14.4	19.2	24.0	28.8	33.6	38.4	43.2	48.0

Table 4: Remifentanil Injection Infusion Rates (ml/h) for a 50 µg/ml Solution

Infusion Rate (µg/kg/min)	**Patient Weight (kg)**							
	30	**40**	**50**	**60**	**70**	**80**	**90**	**100**
0.025	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0
0.05	1.8	2.4	3.0	3.6	4.2	4.8	5.4	6.0
0.075	2.7	3.6	4.5	5.4	6.3	7.2	8.1	9.0
0.1	3.6	4.8	6.0	7.2	8.4	9.6	10.8	12.0
0.15	5.4	7.2	9.0	10.8	12.6	14.4	16.2	18.0
0.2	7.2	9.6	12.0	14.4	16.8	19.2	21.6	24.0
0.25	9.0	12.0	15.0	18.0	21.0	24.0	27.0	30.0
0.5	18.0	24.0	30.0	36.0	42.0	48.0	54.0	60.0
0.75	27.0	36.0	45.0	54.0	63.0	72.0	81.0	90.0
1.0	36.0	48.0	60.0	72.0	84.0	96.0	108.0	120.0
1.25	45.0	60.0	75.0	90.0	105.0	120.0	135.0	150.0
1.5	54.0	72.0	90.0	108.0	126.0	144.0	162.0	180.0
1.75	63.0	84.0	105.0	126.0	147.0	168.0	189.0	210.0
2.0	72.0	96.0	120.0	144.0	168.0	192.0	216.0	240.0

Table 5: Remifentanil Injection Infusion Rates (ml/h) for a 250 µg/ml Solution

Infusion Rate (µg/kg/min)	**Patient Weight (kg)**							
	30	**40**	**50**	**60**	**70**	**80**	**90**	**100**
0.1	0.72	0.96	1.20	1.44	1.68	1.92	2.16	2.40
0.15	1.08	1.44	1.80	2.16	2.52	2.88	3.24	3.60
0.2	1.44	1.92	2.40	2.88	3.36	3.84	4.32	4.80
0.25	1.80	2.40	3.00	3.60	4.20	4.80	5.40	6.00
0.5	3.60	4.80	6.00	7.20	8.40	9.60	10.80	12.00
0.75	5.40	7.20	9.00	10.80	12.60	14.40	16.20	18.00
1.0	7.20	9.60	12.00	14.40	16.80	19.20	21.60	24.00
1.25	9.00	12.00	15.00	18.00	21.00	24.00	27.00	30.00
1.5	10.80	14.40	18.00	21.60	25.20	28.80	32.40	36.00
1.75	12.60	16.80	21.00	25.20	29.40	33.60	37.80	42.00
2.0	14.40	19.20	24.00	28.80	33.60	38.40	43.20	48.00

The following table provides the equivalent blood remifentanil concentration using a TCI approach for various manually-controlled infusion rates at steady state:

Table 6. Remifentanil Blood Concentrations (nanograms/ml) estimated using the Minto (1997) Pharmacokinetic Model in a 70 kg, 170 cm, 40 Year Old Male Patient for Various Manually-Controlled Infusion rates (micrograms/kg/min) at Steady State.

Remifentanil Infusion Rate (micrograms/kg/min)	Remifentanil Blood Concentration (nanograms/ml)
0.05	1.3
0.10	2.6
0.25	6.3
0.40	10.4
0.50	12.6
1.0	25.2
2.0	50.5

MANUFACTURER
LABORATORIO REIG JOFRE S.A.
c/ GRAN CAPITAN n 10
08970 Sant Joan Despi
Barcelona SPAIN

PRODUCT REGISTRATION HOLDER
PAHANG PHARMACY SDN. BHD.
Lot 5979, Jalan Teratai, 5 ½ Mile Off
Jalan Meru, 41050 Klang, Selangor
Malaysia.

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